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Parametric amplification by unbiased Josephson junctions*

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The theory of parametric amplification by a series of unbiased Josephson junctions is presented. The various regimes of operation are explored. This device is in many ways different from the usual varactor parametric amplifier. In particular, a very small pump power is required and a large bandwidth is obtained. Comparison with experimental data is made.

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I. INTRODUCTION

In previous papers,[1,2] we have reported on the preliminary analysis and the observation of parametric amplification in a series of unbiased Josephson junctions. This parametric amplifier is simple to construct, requires very little pump power for high gain, is tunable over a broad frequency range, yields a wide instantaneous bandwidth, and possesses a low noise temperature. These attributes make it an excellent candidate for use as microwave or millimeter-wave amplifier.

In this paper, we develop the theory of this process more extensively. We solve the strong-wave problem, i.e., the nonlinear response of the Josephson element to the strong pump wave, which enables us to display the gain as a function of pump power. We develop a construct, the "infinite nonreentrant gain curve," which allows us to understand and to quantify the various regimes of operation of this device. We can then make a quantitative comparison of the theory with previous 10-GHz experiments. In this analysis, we have included the effects of the important harmonic and parasitic frequencies, and the effects of the "$\cos\phi$" term, the phase-dependent conductivity. Our theory provides a simple explanation of the "glitch" phenomenon as seen in our experiments and reported by others in the literature.

The conventional parmetric amplifier utilizes the nonlinear capacitance of a varactor diode. The present approach uses the nonlinear *inductance* of a Josephson junction as the active element. The current-voltage equation for an ideal Josephson element can be put into the form

$$V=\frac{d}{dt}[L_J(I)I], \tag{1}$$

$$L_J(I)=L_J\frac{\sin^{-1}(I/I_J)}{I/I_J}, \tag{2}$$

$$L_J=\hbar/2eI_J.$$

The ideal Josephson element is then a nonlinear inductor, with characteristic inductance L_J at zero bias current.

We wish to operate with zero bias current. An unbiased Josephson element between identical superconductors is symmetric to a physical inversion. This symmetry operation, which carries $I\to -I$ and $V\to -V$ simultaneously, cannot change the equations which describe the element. This implies $L(I)=L(-I)$. Therefore the nonlinear inductance can depend only upon even powers of the current. [This can also be seen by expanding the arcsin in Eq. (2).] It then follows from Eq. (1) that even harmonics are *absent* in V. In particular, there is no dc rectified or second-harmonic voltages, and no voltages at the sum or difference frequencies.

This symmetry, which is not present in a varactor diode and which is broken by a dc bias, is crucial to understanding the operation of an unbiased Josephson element. In any parametric amplifier, power is supplied at the pump frequency (ω_p), and the system may generate power at the signal frequency (ω_s) and the idler frequency (ω_i). In most conventional parametric amplifiers the primary process is a three-photon process. For instance, an amplifier in which a pump photon decays into a signal photon and an idler photon is described by the frequency relation $\omega_p=\omega_s+\omega_i$. But this process and all other odd-numbered photon processes are not allowed in an unbiased junction by symmetry. The primary process in our case is the four-photon process given by the relation $2\omega_p=\omega_s+\omega_i$. *Two* pump photons decay in our system into a signal photon and an idler photon. The signal and idler frequencies are equally spaced on either side of the pump frequency. The Manley-Rowe relations,

$$P_s/\omega_s=P_i/\omega_i,$$

$$P_p/\omega_p+P_s/\omega_s+P_i/\omega_i=0,$$

indicate that this device acts as a negative resistance amplifier[3] and that the signal and idler share equally (by number) the pump photons which decay by stimulated emission.

We call our device the superconducting unbiased parametric amplifier (SUPARAMP). Its unbiased four-photon mode of operation greatly simplifies the required circuitry. The signal and idler are simply sidebands of the pump; all three frequencies of interest can be handled by the same circuit. Since the sum and difference frequencies are not present, the first important parasitics are near the third-harmonic frequency. If we deliberately short out the third harmonic then these parasitics, the third-harmonic sidebands, are approximately shorted as well. The strongest remaining parasitics are then the fifth-harmonic sidebands. We see that elimination of the parasitics is comparatively simple for the SUPARAMP. Another advantage of the unbiased junction is the absence of electrical leads.

Our active element is chosen to be N identical series-connected junctions in shunt across a transmission line of impedance Z_0 (see Fig. 1). We have not quantitatively accounted for the heterogeneity that must be present in any real set of junctions. A series of N junctions presents an impedance N times larger than a single one of these junctions so that the use of multiple junctions acts as an impedance transformer.

II. THEORY

The equations governing a Josephson junction, in the usual notation, are[4]

$$\frac{d\phi}{dt}=\frac{2e}{\hbar}v, \tag{3}$$

$$J=J_J(v)\sin\phi+[\sigma_0(v)+\sigma_1(v)\cos\phi]v. \tag{4}$$

Assuming J_J, σ_0, and σ_1 to be independent of voltage and assuming a uniform current density, Eq. (4) can be rewritten

$$i=I_J\sin\phi+v/R_J+(\zeta v/R_J)\cos\phi, \tag{5}$$

where $\zeta\equiv\sigma_1/\sigma_0$.

At first we wish to simplify the analysis as much as possible while retaining the essential features. We postpone consideration of the $\cos\phi$ term in Eq. (5) until Sec. II G. We treat the quasiparticle resistance R_J as if it were a separate circuit element, as shown in Fig. 1. The current through the bare Josephson element is then

$$i=I_J\sin\phi. \tag{6}$$

A. Ideal junction response

We wish to examine the response of a junction characterized by Eqs. (3) and (6) to sinusoidal voltages at the pump, signal, and idler frequencies, indexed $k=0$, 1, and 2, respectively:

$$v=\sum_k v_k\cos(\omega_k t+\theta_k). \tag{7}$$

We assume for now that voltages at all other frequencies are short-circuited. We can choose $\theta_0=0$, so that θ_1 and θ_2 signify the phases of the signal and idler voltages with respect to the pump voltage. In this section, we find the currents induced by these applied voltages. Integrating Eq. (3) and using Eq. (7), we see that the junction phase difference, ϕ, is given by

$$\phi(t)=\delta+\sum_k\phi_k(t), \tag{8}$$

$$\phi_k(t)=\alpha_k\sin(\omega_k t+\theta_k),$$

$$\alpha_k=2ev_k/\hbar\omega_k.$$

In an unbiased junction, $\delta=0$. Requiring α_1, $\alpha_2\ll 1$ (α_0 is not restricted), we can derive *linear* equations relating the signal and idler waves. With this restriction, we can substitute Eq. (8) into Eq. (6) and find

$$i=I_J\sin\phi_0(t)+I_J[\phi_1(t)+\phi_2(t)]\cos\phi_0(t).$$

We must now isolate the components of this current which flow at the three allowed frequencies. We use the equality

$$\exp(j\alpha_0\sin\omega_0 t)=\sum_{k=-\infty}^{\infty}J_k(\alpha_0)\exp(jk\omega_0 t), \tag{9}$$

and the frequency relation $2\omega_0=\omega_1+\omega_2$. The amplitudes of the currents which oscillate at the three allowed frequencies are, in complex notation,[5]

$$i_0=-2jI_JJ_1(\alpha_0), \tag{10}$$

$$i_1=-j\frac{2eI_J}{\hbar}\left(J_0(\alpha_0)\frac{v_1}{\omega_1}-J_2(\alpha_0)\frac{v_2^*}{\omega_2}\right), \tag{11a}$$

$$i_2=-j\frac{2eI_J}{\hbar}\left(J_0(\alpha_0)\frac{v_2}{\omega_2}-J_2(\alpha_0)\frac{v_1^*}{\omega_1}\right). \tag{11b}$$

α_0 is real. The voltages and currents, however, are complex amplitudes, containing the phase with respect to the pump voltage. Equation (10), describing the response of the junction to the strong pump wave, is nonlinear. Equations (11a) and (11b), coupling the two weak waves, are linear. Thus Eqs. (11a) and (11b) can be expressed in matrix form:

$$\begin{bmatrix} i_1 \\ i_2^* \end{bmatrix}=\begin{bmatrix} Y_{11} & Y_{12}^* \\ Y_{21} & Y_{22}^* \end{bmatrix}\begin{bmatrix} v_1 \\ v_2^* \end{bmatrix}, \tag{12}$$

where

$$Y_{11}=-j\frac{J_0(\alpha_0)}{L_J\omega_1},\quad Y_{12}=-j\frac{J_2(\alpha_0)}{L_J\omega_2},$$

$$Y_{21}=-j\frac{J_2(\alpha_0)}{L_J\omega_1},\quad Y_{22}=-j\frac{J_0(\alpha_0)}{L_J\omega_2}.$$

We shall call Y_{11} and Y_{22} *self-coupling* and Y_{12} and Y_{21} *cross-coupling* susceptances.

B. Equivalent circuit

Figure 1 represents the circuit model used for the theoretical calculations. N closely spaced series-connected Josephson junctions are placed in shunt across a transmission line. The junctions are assumed to be identical. Each junction is completely characterized by the critical current, I_J, and by the quasiparticle resistance, $R_J=1/G_J$. For the strong wave at ω_0, Y in Fig. 1 is the nonlinear admittance implicitly given by Eq. (10). For the weak waves at ω_1 and ω_2 the admittance Y is the linear admittance matrix of Eq. (12). The transmission line has characteristic impedance $Z_0=1/G_0$ and is terminated by Y_T which is imaginary.

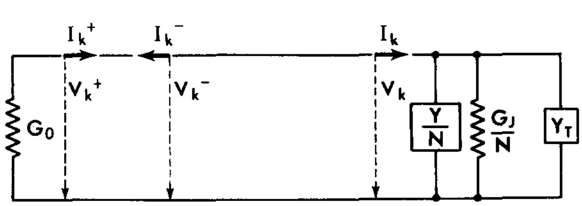

FIG. 1. Simplified circuit of a reflection amplifier. A series of N Josephson junctions is represented by the parallel combination of the admittance Y/N arising from the Josephson inductance, and the quasiparticle conductance G_J/N. The subscript k represents in turn all allowed frequencies.

For now we assume Z_0 and Y_T do not depend upon frequency.

The traveling waves represented in Fig. 1 must obey

$$I_k^{\pm} = G_0 V_k^{\pm},$$

where the plus and minus signs respectively denote the incident and reflected waves. The resultant current and voltage at the end of the transmission line are

$$I_k = I_k^+ - I_k^-$$

and

$$V_k = V_k^+ + V_k^-,$$

where $V_k = N v_k$ is the total voltage across N identical junctions and $I_k = i_k + G_J v_k + Y_T V_k$ is the total current through the bare Josephson elements, the junction resistances, and the termination. The incident power P_k^+ and reflected power P_k^- at frequency ω_k are

$$P_k^{\pm} = \tfrac{1}{2} G_0 |V_k^{\pm}|^2 = (8G_0)^{-1} |I_k \pm G_0 V_k|^2. \tag{13}$$

C. Strong-wave problem

The assumption $\alpha_1, \alpha_2 \ll 1$ separates this calculation into two parts: the strong-wave problem and the weak-wave problem. In the strong-wave problem, the junctions are driven by a large-amplitude incident pump wave at ω_0 and respond nonlinearly. This can be analyzed without mention of the weak waves at ω_1 and ω_2. The weak waves will be treated as a perturbation upon the strong pump wave.

Using Eq. (10), the total pump current is

$$I_0 = -2jI_J J_1(\alpha_0) + G_J V_0/N + Y_T V_0.$$

It follows from Eq. (13) that the incident and reflected pump powers are

$$P_0^{\pm} = \tfrac{1}{8} I_J^2 Z_0 \{[2J_1(\alpha_0) - g\xi b_T \alpha_0]^2 + (1 \pm g)^2 \xi^2 \alpha_0^2\}, \tag{14}$$

where $Z_0 = 1/G_0$ and we have introduced the dimensionless parameters

$$g = \frac{NG_0}{G_J} = \frac{NR_J}{Z_0},$$

$$\xi = \omega_0 L_J G_J = \frac{\omega_0 L_J}{R_J} = \frac{\hbar\omega_0}{2eI_J R_J},$$

$$jb_T = \frac{Y_T}{G_0}.$$

The parameter g compares the external load with the internal load, and therefore scales with the number of junctions, N. The parameter ξ describes the quality of the individual junction, independent of N.

The dimensionless parameter

$$\alpha_0 = \frac{2ev_0}{\hbar\omega_0} = \frac{2eV_0}{N\hbar\omega_0}$$

represents the pump voltage amplitude across the individual junction. It is the variable which governs the state of the system. However, it is an *internal* variable not subject to the experimenter's direct control. The incident pump power is not directly proportional to α_0^2. In fact, the appearance of $J_1(\alpha_0)$ in Eq. (14) implies that the voltage across the Josephson elements can be a multiply-valued function of the incident pump power. As the incident pump power is monotonically varied, the internal variable α_0 (i.e., the pump voltage amplitude) can discontinuously jump, causing "glitches" and hysteresis in the experimental observables. This is discussed at length below.

D. Weak-wave problem

From Eq. (13) we see that the signal power gain Γ is

$$\Gamma \equiv \frac{P_1^-}{P_1^+} = \left|\frac{Y_1 - G_0}{Y_1 + G_0}\right|^2, \tag{15}$$

where we have defined the input admittance $Y_1 \equiv I_1/V_1$. The fact that there is no power incident at the idler frequency, $P_2^+ = 0$, implies $I_2 = -G_0 V_2$. We then define the conversion ratio

$$C \equiv \frac{P_2^-}{P_1^-} = 4G_0^2 \left|\frac{V_2^*/V_1}{Y_1 - G_0}\right|^2.$$

To evaluate these quantities we need to determine Y_1 and V_2^*/V_1. Using Eq. (12), the current-voltage equations for the signal and the idler are

$$\begin{aligned} I_1 &= Y_1 V_1 \\ &= \frac{Y_{11} V_1}{N} + \frac{Y_{12}^* V_2^*}{N} + \frac{G_J V_1}{N} + Y_T V_1, \end{aligned} \tag{16a}$$

$$\begin{aligned} I_2^* &= -G_0 V_2^* \\ &= \frac{Y_{22}^* V_2^*}{N} + \frac{Y_{21} V_1}{N} + \frac{G_J V_2^*}{N} + Y_T^* V_2^*. \end{aligned} \tag{16b}$$

Equation (16b) directly gives V_2^*/V_1. Substituting this into Eq. (16a) gives Y_1:

$$Y_1 = \frac{Y_{11}}{N} + Y_T + \frac{G_J}{N} - \frac{Y_{12}^* Y_{21}/N^2}{Y_{22}^*/N + Y_T^* + G_J/N + G_0}. \tag{17}$$

From Eqs. (15), (17), and (12), it follows that the signal gain for the degenerate case, $\omega_1 = \omega_0 = \omega_2$, is

$$\begin{aligned} \Gamma = (&\{\xi^2 - g^2\xi^2 + [J_0(\alpha_0) - g\xi b_T]^2 - J_2(\alpha_0)^2\}^2 \\ &+ 4g^2\xi^2[J_0(\alpha_0) - g\xi b_T]^2)\{\xi^2(1+g)^2 \\ &+ [J_0(\alpha_0) - g\xi b_T]^2 - J_2(\alpha_0)^2\}^{-2}. \end{aligned} \tag{18}$$

E. Properties of the solution

Some characteristics of this solution should be emphasized:

(i) α_0, the physical state variable of the system, is the argument of the Bessel functions in these solutions. It will be the independent variable in all computations. (For the remainder of this paper the argument of all Bessel functions, α_0, will be omitted.)

(ii) Note that the incident and reflected pump powers (except for the coefficient $\frac{1}{8} I_J^2 Z_0$) and the degenerate gain are solely functions of the dimensionless parameters α_0, ξ, g, and b_T.

(iii) This device is a negative resistance amplifier. Although the negative conductance depends upon the details of the termination, we can determine its magnitude by examination of Eq. (17) for the special case in which $\omega_0 = \omega_1 = \omega_2$ and in which we choose $b_T = J_0/g\xi$ to set $\mathrm{Im}\, Y_1 = 0$. It then follows that

$$\mathrm{Re}Y_1 = \frac{G_J}{N} - \frac{J_2^2}{(N\omega_0 L_J)^2(G_0 + G_J/N)}. \tag{19}$$

The last term is the negative conductance of the bare Josephson element.

(iv) Inspection of Eq. (17) leads to the following general theorem: Any degenerate ($\omega_1 = \omega_2$) parametric amplifier consisting of a nonlinear element in shunt across a transmission line with arbitrary termination has infinite gain if and only if $\mathrm{Re}Y_1 = -G_0$. It is remarkable that the vanishing of $\mathrm{Re}(Y_1 + G_0)$ implies that $\mathrm{Im}(Y_1 + G_0)$ vanishes as well. For the Josephson nonlinearity [Eq. (12)], the condition $\mathrm{Re}Y_1 = -G_0$ can be written

$$D \equiv \xi^2(1+g)^2 + (J_0 - g\xi b_T)^2 - J_2^2 = 0. \tag{20}$$

Note that D^2 is the denominator of Γ [Eq. (18)], verifying the theorem for our system. Equation (20) implies that $\xi(1+g) \leq \max(J_2)$ is a necessary (but not sufficient) condition for infinite gain.

(v) From inspection of Eq. (20) it is seen that the gain can be infinite without choosing b_T such that $J_0 - g\xi b_T = 0$. In other words, it is not necessary to resonate out the self-coupling susceptance to achieve high gain.

(vi) From Eq. (14) we find the important relation

$$\frac{dP_0^+}{d(\alpha_0^2)} = \tfrac{1}{8} I_J^2 Z_0 D, \tag{21}$$

where D^2 is the denominator of the degenerate gain, Eq. (18). In particular, $dP_0^+/d\alpha_0 = 0$ for infinite gain. Equation (21) forms a unique connection between the strong- and weak-wave processes.

F. Harmonic problem

We now consider power flow at frequencies other than the signal, pump, and idler. Power can flow at the nth harmonic of the pump frequency, $n\omega_0$, and its sidebands $n\omega_0 \pm (\omega_0 - \omega_1)$, where n is an odd integer. Higher-order sidebands, $n\omega_0 \pm m(\omega_0 - \omega_1)$ where $m > 1$, are vanishingly small in the weak-wave approximation, α_1, $\alpha_2 \ll 1$. We have already seen that the even-numbered harmonics, including the frequency component $\omega_0 - \omega_1$, cannot exist due to the symmetry of the unbiased junction. To simplify the problem, we choose the termination to be a short circuit at the third harmonic $3\omega_0$. This ensures that the third-harmonic voltage and the most important parasitic voltages, at $3\omega_0 \pm (\omega_0 - \omega_1)$, are small. At the moderate pump powers that we will find optimum, fifth-harmonic voltages are also small. The seventh and higher harmonics are neglected in our calculations. In order to determine the bandwidth of the device, the requirement of degeneracy is removed: the frequency separation $\omega_0 - \omega_1$ is varied. The dispersion of the transmission line and the additional frequency dependence of the termination can also, in principle, be arbitrary.

Again we separate the calculation into two parts: the strong-wave problem and the weak-wave problem. The strong-wave problem now involves determining the voltages v_{30} at $3\omega_0$ and v_{50} at $5\omega_0$ in terms of α_0, in addition to finding the expression for $P_0^{\pm}(\alpha_0)$. Let us define the dimensionless harmonic voltages $\alpha_{30} \equiv 2ev_{30}/3\hbar\omega_0$ and $\alpha_{50} \equiv 2ev_{50}/5\hbar\omega_0$ in analogy with Eq. (8). We require α_{30} and α_{50} to be small in order to linearize the current-voltage equations in these quantities. To ensure this, the termination Y_T is chosen to be a short at $3\omega_0$. We have the freedom to vary Y_T from a short at $3\omega_0$, as long as α_{30} and α_{50} are both small. (α_{30}, $\alpha_{50} < 0.3$ is sufficient.) In the Appendix, the linearized current-voltage equations and the circuit equations for the third and fifth harmonics are constructed in order to obtain α_{30} and α_{50} as functions of α_0. Given α_0, α_{30}, and α_{50}, we can proceed to solve the weak-wave problem. The signal gain, the idler conversion ratio, and the conversion ratios at the frequencies $n\omega_0 \pm (\omega_0 - \omega_1)$, $n = 3$ and 5, are calculated in the Appendix. Note that the weak-wave admittance matrix is now 6×6 rather than 2×2.

All of the theoretical curves in this paper are computed using this harmonic analysis. We choose, as standard, a microstrip transmission line, with the junctions terminated by a shunt section of open line of length L. The line is nondispersive, and the specific frequency dependence of the termination is $b_T(\omega) = \tan(\omega L/v_p)$, where v_p is the phase velocity. Our standard termination length is chosen to be $L = \frac{1}{12}\lambda_g$ (implying $b_T = 1/\sqrt{3}$) unless otherwise noted. This is the shortest length L which is a short at $3\omega_0$.

G. Cosϕ term

The $\cos\phi$ term was theoretically predicted specifically for tunnel junctions.[4] The effects of this term have been experimentally observed in point contacts[6] as well as tunnel junctions.[7] Although the junctions used in the experiments described below are thin-film constriction microbridges, we expect a phase-dependent conductivity to exist in these junctions as well.

It is a simple matter to include the effects of the $\cos\phi$ term in the above calculations. If we ignore the linear quasiparticle current v/R_J, the rest of Eq. (5) can be written

$$i = I_J\left(1 + \frac{\zeta L_J}{R_J}\frac{d}{dt}\right)\sin\phi.$$

We can Fourier analyze the supercurrent

$$I_J \sin\phi = \sum_\omega a_\omega \exp(j\omega t).$$

The current at frequency ω is then

$$i_\omega = (1 + j\omega\tau)a_\omega,$$

where $\tau \equiv \zeta L_J/R_J = \zeta\xi/\omega_0$. We find from this a simple prescription for including the effects of the $\cos\phi$ term: the current flowing at frequency ω through the bare Josephson elements must be multiplied by the factor $1 + j\omega\tau$.

Including the $\cos\phi$ term in the nonharmonic calculation, the incident and reflected pump powers become

$$P_0^{\pm} = \frac{I_J^2 Z_0}{8}\xi^2\alpha_0^2\left[\left(\frac{2J_1}{\alpha_0\xi} - gb_T\right)^2 + \left(\frac{2J_1\zeta}{\alpha_0} + 1 \pm g\right)^2\right] \tag{22}$$

and the square root of the denominator of the gain becomes

$$D = \xi^2(1 + g + \zeta J_0)^2 + (J_0 - g\xi b_T)^2 - (1 + \zeta^2\xi^2)J_2^2. \tag{23}$$

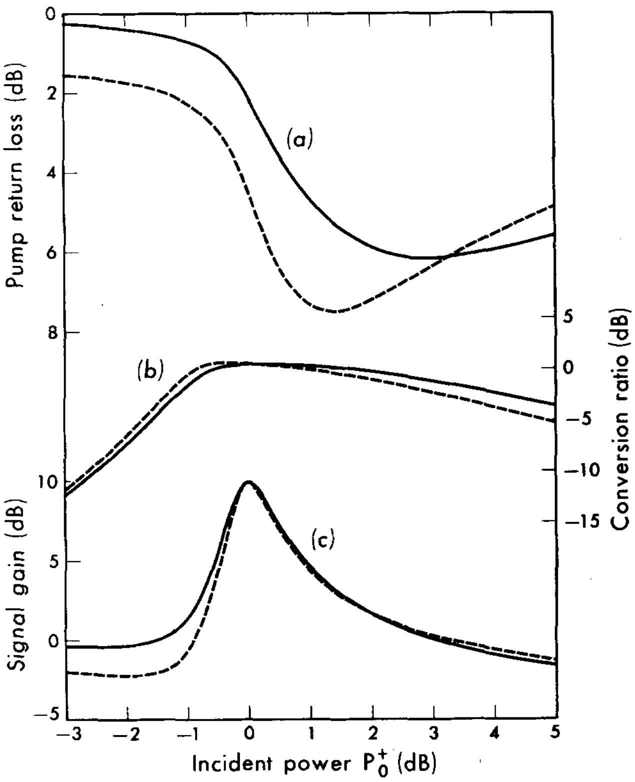

FIG. 2. (a) Pump return loss, (b) conversion ratio, and (c) signal gain plotted against incident pump power. The standard termination is assumed. $g=2.5$, $\xi=0.143$, and $\zeta=-1$ for the solid curves; $g=2.5$, $\xi=0.139$, and $\zeta=0$ for the dashed curves.

Note that Eq. (21) relating P_0^+ and D is still exactly true. In the limit of very high gain, the gain approaches

$$\Gamma \rightarrow \frac{4g^2\xi^2(1+\zeta^2\xi^2)J_2^2}{D^2} .$$

Although there is some uncertainty theoretically as to its sign,[8] experiments indicate that ζ is about equal to -1, at least at zero temperature. Calculations have been carried out for $\zeta=-1$ and $\zeta=0$. We find that the effects of the $\cos\phi$ term are not clearly distinguishable in the experiments we have performed.

III. DISCUSSION

In this section, we discuss the theoretical results and examine the implications of varying the important parameters of the theory. Unless otherwise noted, the harmonics and the $\cos\phi$ term are included.

A. Basic results

Figures 2–5 are typical results of these calculations. These results are displayed as a function of the incident pump power, P_0^+, for comparison with experiment. The parameters g and ξ used in computing these curves are given in the figure captions. These parameters have been chosen to yield 10 dB peak gain. We assume our standard termination, with zero frequency shift of the signal from the pump and $\zeta=-1$. The dashed curves in Figs. 2 and 4 are for $\zeta=0$ (with ξ slightly adjusted to maintain 10 dB peak gain). P_0^+ is given in dB relative to the power for peak gain. The absolute value of P_0^+ depends upon one's choice of I_J and Z_0.

Figure 2(a) shows the behavior of the pump return loss, defined P_0^+/P_0^-, as a function of P_0^+ for a particular set of parameters. The transition from linear to nonlinear behavior as P_0^+ is increased is evident. If we were to decrease ξ (for example, by decreasing the operating temperature), the knee in the return loss would become steeper until, at a certain value of ξ, the slope of the return-loss—vs—P_0^+ curve would be infinite at one point. As ξ is decreased further, the return-loss curve becomes "reentrant", as shown by Fig. 3(a). This curve represents a multiply-valued function. Once the slope in Fig. 3(a) becomes infinite with increasing P_0^+, the return loss jumps discontinuously along the dashed arrow in Fig. 3(a), reaching the lower stable curve. This is seen as a downward "glitch" in the return loss as P_0^+ is increased. There is a corresponding upwards glitch in return loss as P_0^+ is decreased. As seen from Fig. 3(a), these two glitches do not occur at the same value of P_0^+: the curve is hysteretic. The central region of this curve is not physically accessible. If ξ is decreased still further, the separation of the two glitches increases.

Figures 2(c) and 3(c) show the signal gain as a function of the incident pump power for the same parameters as in Figs. 2(a) and 3(a), respectively. In Fig. 2(c), the

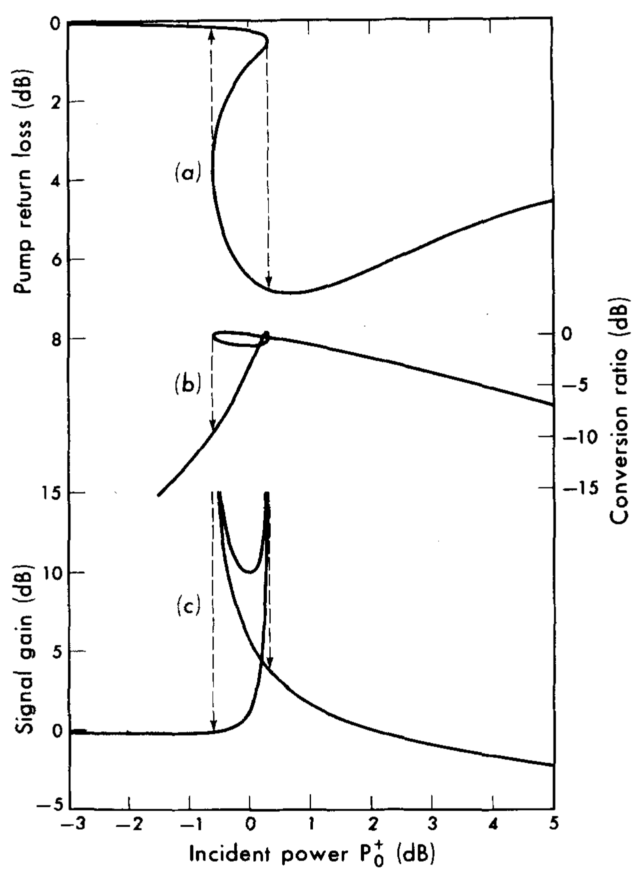

FIG. 3. (a) Pump return loss, (b) conversion ratio, and (c) signal gain plotted against incident pump power. The standard termination is assumed. $g=2.5$, $\xi=0.091$, and $\zeta=-1$. Dotted arrows indicate glitches as pump power is swept.

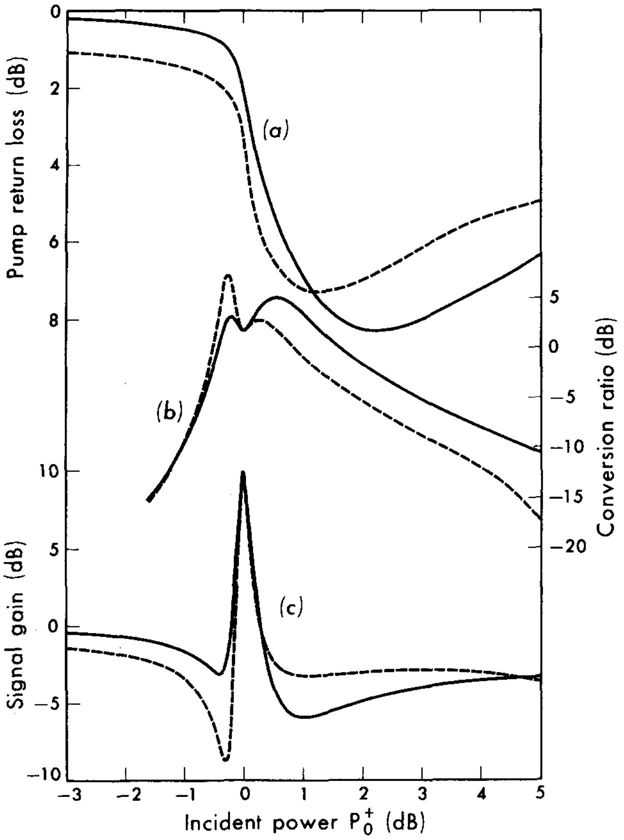

FIG. 4. (a) Pump return loss, (b) conversion ratio, and (c) signal gain plotted against incident pump power. The standard termination is assumed. $g=0.4$, $\xi=0.351$, and $\zeta=-1$ for the solid curves; $g=0.4$, $\xi=0.329$, and $\zeta=0$ for the dashed curves.

signal gain has a finite peak as a function of P_0^+. If ξ is decreased from its value in Fig. 2(c), the gain peak grows in height and narrows in width until, at a certain value of ξ, it becomes an infinitely narrow infinite gain peak. This occurs at the same value of ξ and of P_0^+ as the first occurrence of infinite slope in the return loss mentioned above. As ξ is further decreased, the gain-vs-P_0^+ curve also becomes reentrant. Again there are glitches in the gain as P_0^+ is monotonically varied [illustrated by the dashed arrows in Fig. 3(c)]; again the central region of this curve is physically inaccessible. The glitches in gain exactly correspond to the glitches in return loss. Figures 2(b) and 3(b) show the idler conversion ratio as a function of incident pump power for the same parameters as in Figs. 2(a) and 3(a), respectively.

Figures 2 and 3 were computed for $g=2.5$, chosen to illustrate the case $g>1$. Figures 4 and 5 illustrate corresponding theoretical curves for the case $g<1$. Again, ξ is chosen to give a peak gain of 10 dB. Note that the curves in Figs. 4 and 5 are similar to those of Figs. 2 and 3, respectively, except that the corresponding features occur over a narrower range of incident pump power, and that these features are more pronounced.

Reentrance in the curves of Figs. 3 and 5 and consequent glitching can be understood in terms of the behavior of the voltage across the junction, discussed in Sec. II C. Due to the peculiar nonlinearity of the Josephson junction, the pump voltage amplitude can be a multiply-valued function of the incident pump power. When this occurs, the gain, the conversion ratio, and the return loss, which are functions of α_0, will be multiply-valued functions of the incident pump power: the curves of these functions will be reentrant.

The central physically inaccessible region of these curves begins and ends at the points where $dP_0^+/d\alpha_0=0$. We see from Eq. (21) that these are the very points where the gain is infinite (at least for the case of perfectly shorted higher harmonics). Thus, in Figs. 3(c) and 5(c), glitching occurs at the two infinite gain peaks.

Glitches in reflected power have been reported in experiments similar to ours.[9,10] In these experiments, a thin superconducting film with no apparent Josephson junctions was placed in a high-Q cavity. Although a detailed analysis of these experiments is not possible, we believe their glitches result from a similar process. In reports of other experiments[11,12] no glitches are mentioned.

As α_0 is increased past the gain peak, the Bessel functions oscillate. This causes further peaks in the gain at higher values of pump power. Computer studies show that these higher α_0 peaks are of less practical interest and reveal no new concepts or behavior. However, there can be multiple glitches at higher pump powers similar to those reported in Refs. 9 and 10.

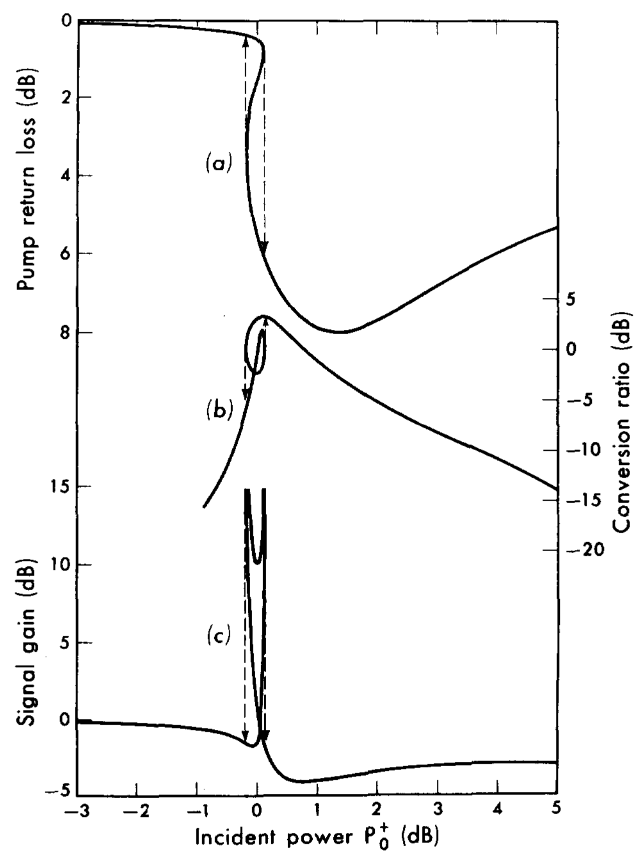

FIG. 5. (a) Pump return loss, (b) conversion ratio, and (c) signal gain plotted against incident pump power. The standard termination is assumed. $g=0.4$, $\xi=0.282$, and $\zeta=-1$. Dotted arrows indicate glitches as pump power is swept.

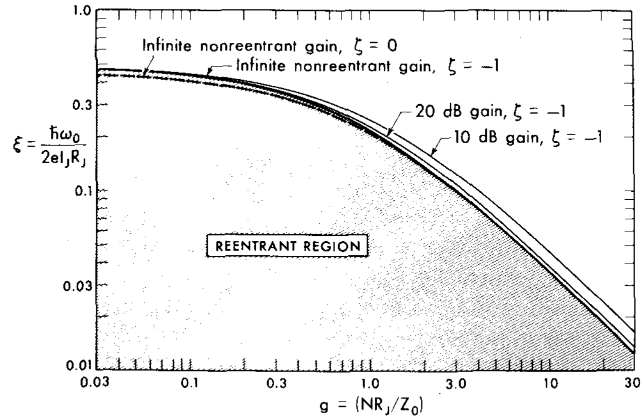

FIG. 6. Locus of points (g, ξ) giving infinite nonreentrant gain for $\zeta = 0$; and infinite nonreentrant gain, 20 dB peak gain, and 10 dB peak gain all for $\xi = -1$. The standard termination is assumed.

B. Infinite nonreentrant gain

The results discussed in Sec. III A naturally divide into two regimes: reentrant and nonreentrant. In particular, the gain curves can either be reentrant or have finite peak gain. These curves, plotted as functions of P_0^+, depend upon the parameters g, ξ, ζ, and b_T. For a given value of g, fixing ζ and b_T, we saw that as ξ is decreased, the peak gain grows until it becomes infinite. For smaller ξ, the gain curve is reentrant. Thus there is an infinite but not reentrant gain curve which forms the boundary between the two regimes. The point of infinite gain on this particular curve is given for the nonharmonic degenerate case by the two conditions:

$$D(\alpha_0, g, \xi; \zeta, b_T) = 0, \tag{24}$$

$$\frac{\partial D}{\partial \alpha_0}(\alpha_0, g, \xi; \zeta, b_T) = 0, \tag{25}$$

where D is given by Eq. (23). The second condition can be written

$$[g\xi b_T - \zeta\xi^2(1+g)]J_1 + 2(1+\zeta^2\xi^2)(J_2^2 - J_1^2)/\alpha_0 = 0.$$

Clearly, this procedure is valid whatever the choice of g. Therefore, assuming fixed values for ζ and b_T we have two equations in three unknowns: α_0, g, and ξ. Eliminating α_0 from these two equations generates a curve in (g, ξ) space parametrized by α_0. This curve represents the complete set of values of (g, ξ) for which infinite but not reentrant gain occurs. We call this the *infinite nonreentrant gain* (ING) curve. In Fig. 6, the ING curve is plotted for $\zeta = -1$, $b_T = 1/\sqrt{3}$ and for $\zeta = 0$, $b_T = 1/\sqrt{3}$. The form of these curves depends weakly upon b_T for $|b_T| < 1$. α_0 varies very little along these curves. (For the entire $\zeta = -1$ curve, the range α_0 is $2.25 < \alpha_0 < 2.31$.)

To physically understand this procedure for obtaining the ING curve, let us consider an amplifier with a given g (i.e., a given N, R_J, and Z_0). At the critical temperature of the superconductor, ξ is infinite and there is no gain. As we lower the temperature of the amplifier, ξ decreases and gain appears. At a certain temperature the gain peak becomes infinite. The value of ξ at this temperature determines a point in (g, ξ) space corresponding to infinite but nonreentrant gain. Below this temperature the amplifier will exhibit glitching.

In general, the parameters N, R_J, I_J, ω_0, and Z_0 determine an operating point in (g, ξ) space. If this point lies below the ING curve the gain will be reentrant, as in Figs. 3 and 5. If it lies above the curve the peak gain will be finite and nonreentrant, as in Figs. 2 and 4. From a practical point of view, high stable gain is preferred. Although reentrant gain curves permit high and indeed infinite gain, the gain is sensitive to small changes in incident pump power. Finite nonreentrant gain curves permit stable operation at peak gain. Therefore, the preferred values of (g, ξ) lie close to but slightly above the ING curve. This is illustrated by the curves of (g, ξ) for 20 and 10 dB peak gain which are shown in Fig. 6 ($\zeta = -1$, harmonics included with our standard termination). Note that if we require, for instance, 10 dB peak gain, then given g the choice of ξ is no longer arbitrary.

We can arrive at the form of the ING curve with some simple intuitive considerations. For simplicity let us ignore the termination, the Josephson self-coupling terms, and the $\cos\phi$ term. From Eq. (19), we can see that the magnitude of the negative resistance of the bare Josephson elements must then be $(N\omega_0 L_J/J_2)^2(G_0 + G_J/N)$. Infinite gain occurs whenever the magnitude of the negative resistance is equal to the positive resistance, which is the parallel combination of NR_J and Z_0. In dimensionless parameters, the simplified ING curve is thus characterized by $\xi(1+g) = \max(J_2)$, where $\max(J_2)$ is the maximum value of J_2 (≈ 0.5). In spite of the assumptions we have made, this simplified ING curve very closely approximates the ING curves of Fig. 6.

This simple model suggests that two important limits be considered. If $g \ll 1$, we can ignore Z_0 with respect to NR_J. The ratio $\omega_0 L_J/R_J$ (or ξ) must then be adjusted to give high gain. This implies that, in the limit $g \ll 1$, ξ approaches a constant (≈ 0.5). But once this high gain is achieved, only a small fraction of the amplified signal power is delivered to the external load Z_0 relative to the internal load R_J. In the opposite limit, $g \gg 1$, we can ignore NR_J with respect to Z_0. Now the ratio $N\omega_0 L_J/Z_0$ (or $g\xi$) must be adjusted to give high gain, implying that in this limit the product $g\xi$ approaches a constant (again ≈ 0.5). Most of the amplified signal power is delivered to the external load. This limit is the more desirable one for reasons that will become apparent below.

The defining equations of the ING curve, Eqs. (24) and (25), are significant for the strong-wave properties of the solution as well as for the weak-wave properties. This is evident from Eq. (21). We can calculate the value of incident pump power that will give infinite nonreentrant gain. Equation (22), with the conditions Eqs. (24) and (25), becomes

$$P_0^+(\text{ING}) = \tfrac{1}{2} I_J^2 Z_0 [(1+\zeta^2\xi^2)\alpha_0 J_2^3/J_1]. \tag{26}$$

The factor in brackets is roughly constant along the entire ING curve. The necessary pump power indicated by this equation is very low compared to other parametric

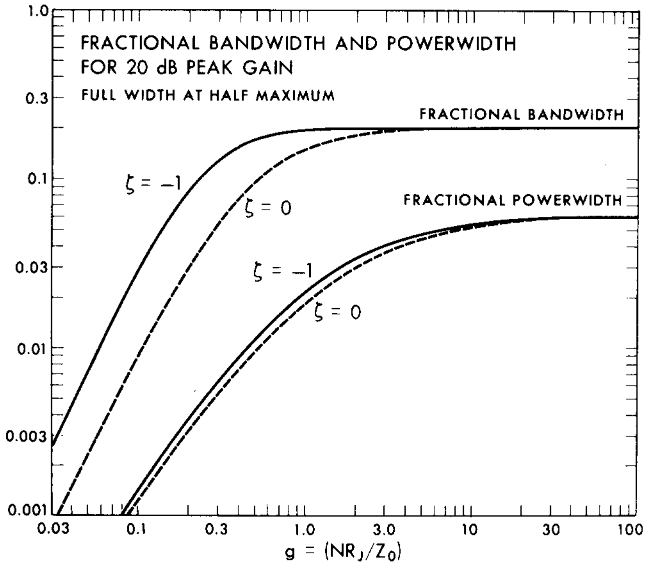

FIG. 7. FWHM fractional bandwidth and FWHM fractional power width for 20 dB peak gain, plotted as a function of g. $\zeta=-1$ for the solid curves and $\zeta=0$ for the dashed curves. The standard termination is assumed.

amplifiers. For example, a set of junctions with critical current 100 μA in a 50-Ω transmission line requires a pump power of ≈ 0.1 μW for high gain. Conversely, we can use Eq. (26) as a measure of I_J.

From Eq. (22), we find the pump power loss

$$P_0^+ - P_0^- = \left(\frac{\hbar\omega_0}{2e}\right)^2 \frac{N}{2R_J}\alpha_0^2\left(1+\frac{2J_1\zeta}{\alpha_0}\right).$$

This can be used to measure N/R_J.

C. Bandwidth and power width

Compared to other reflection parametric amplifiers, the instantaneous bandwidth of the SUPARAMP is very large. In other parametric amplifiers, the self-coupling is always stronger than the cross-coupling.[13] In order to have high gain in these devices it is therefore necessary to resonate out the self-coupling reactance. This allows the cross-coupling, and hence the negative resistance, to be predominant. In the SUPARAMP, however, no such resonance is necessary for high gain. The self-coupling term is automatically smaller than the cross-coupling term [see Eq. (12)], because J_0 is small at the operating point, where J_2 is near maximum. Since we can achieve high gain without resonance, the bandwidth is also automatically large.

The gain-bandwidth (FWHM) product for high gain approaches a constant,

$$\Gamma_c^{1/2}(\Delta\omega/\omega_0) \to \text{const}, \tag{27}$$

where the constant depends upon the operating parameters approached in the limit of infinite gain. This limit can be approached by the variation of any of the parameters. Γ_c denotes the gain at band center. Γ_c is defined for any value of P_0^+; it is not necessarily the peak gain encountered in the gain-vs-P_0^+ curve. An analytic expression for the contant in Eq. (27), ignoring harmonics, is

$$\frac{2g(1+\zeta^2\xi^2)^{1/2}J_2}{|(1+g+\zeta J_0)[J_0+g\xi\omega_0 b_T'(\omega_0)]-\xi J_2^2|}, \tag{28}$$

where $b_T'(\omega_0)$ is the derivative of b_T with respect to frequency, evaluated at ω_0. Note that in the overcoupled ($g \gg 1$) limit and for $b_T'=0$ this constant is $|J_2/J_0|$, the ratio of the cross-coupling to the self-coupling susceptances.[1]

In Fig. 7, the fractional bandwidth (FWHM) for 20 dB peak gain resulting from the harmonic calculation with our standard termination is plotted as a function of g. Comparison of these results with Eqs. (27) and (28) shows that the harmonics are important in limiting the bandwidth for $g<1$, but have little effect for $g>1$. The tuning stub is designed to exactly short out the third harmonic, and will approximately short out the parasitics over a broad frequency range. However as g becomes smaller, the third-harmonic short becomes progressively less effective in preventing power flow through the quasiparticle resistance R_J at the parasitic frequencies. This limits the bandwidth for $g<1$. No such limitation exists for $g>1$.

Since the (nonreentrant) degenerate gain goes through a smooth peak as P_0^+ is increased, we can define a "power width" analogous to the bandwidth. The power width, ΔP, is defined as the separation in incident pump power of the half-maximum gain points. As the gain increases, ΔP becomes quite narrow. There exists a constant gain—power-width product analogous to the gain-bandwidth product. This expression is

$$\Gamma_p^{3/4}(\Delta P/P_0^+) \to \text{const}, \tag{29}$$

where Γ_p denotes the peak gain. The constant depends upon the operating point on the ING curve approached in the limit of infinite peak gain. We explain this expression as follows. We see from Eq. (18) that $\Gamma^{1/2}$, rather than Γ, is Lorentzian in small excursions of α_0 about a high gain peak. As with any Lorentzian function, we can write an expression analogous to the familiar gain-bandwidth product. In this case $\Gamma_p^{1/4}\Delta\alpha_0=\text{const}$, where $\Delta\alpha_0$ is the width of the gain peak in α_0. But Eq. (21) implies that $\Delta\alpha_0$ is proportional to $\Delta P\Gamma^{1/2}$ for high gain. Combining these gives Eq. (29). Computer results verify this prediction. An analytic expression for the constant in Eq. (29), ignoring harmonics is

$$4.14\frac{J_2}{\alpha_0}\frac{(\alpha_0 J_1/J_2-1)^{3/4}}{(3J_1^2-3J_0J_2-2J_2^2)^{1/2}}(1+\zeta^2\xi^2)^{3/4} \times\left|\frac{g}{1+g+\zeta g\xi b_T}\right|^{3/2}. \tag{30}$$

The fractional power width for 20 dB peak gain is plotted in Fig. 7 as a function of g, taking into account the harmonics with our standard termination. Comparison of these results with Eqs. (29) and (30) shows that the harmonics have little effect on the power width.

From Eqs. (28) and (30) we see that both the bandwidth and power width decrease as g becomes small. In this limit, most of the amplified signal power is delivered to the internal load R_J rather than the external load Z_0. For large g both the bandwidth and the power width are independent of g since the internal load is negligible. We have seen that the harmonics further limit the bandwidth for small g. In order to have large bandwidth and power width, it is clear that we wish to operate on the large g side of the ING curve. Multiple junctions are a convenient way of achieving this.

We have implicitly assumed that the incident pump power is unaffected by the process of amplification. In fact, the amplified signal (and idler) power must be drawn from the pump. If this amplified power is comparable to the power width, the analysis of this paper no longer holds and saturation is to be expected. To prevent thermal noise, of power $kT\Delta\nu$, where $\Delta\nu$ is the bandwidth of the amplifier, from saturating the amplifier, we must have $2\Gamma_p kT\Delta\nu < \frac{1}{2}\Delta P$. Thus we can define a saturation temperature,

$$T_{\mathrm{sat}} = \frac{\Delta P/2}{2\Gamma_p k\Delta\nu}.$$

This is the largest thermal signal (or noise background) temperature that can be tolerated without saturation. $\Delta\nu$ and ΔP can be expressed in terms of the gain by use of Eqs. (27) and (29) and Fig. 7. P_0^+ is found from Eq. (26) with I_J determined by the product $g\xi$ evaluated along the ING curve, Fig. 6. We find

$$T_{\mathrm{sat}} = 13.6 N^2 \nu_{10} \Gamma_p^{-5/4} F, \tag{31}$$

where ν_{10} is the operating frequency in units of 10 GHz. F is a function which depends upon the operating point near the ING curve: for large g, $F=1$; in the limit of small g, $F \to 0.045 g^{-5/2}$. F becomes significantly larger than unity only for very small g.

Until now, the use of multiple junctions has seemed to be only a convenient means of compensating for small R_J. From Eq. (31), however, we see that multiple junctions must be used in the SUPARAMP to prevent saturation. A single high-resistance junction operating at 10 dB peak gain at 10 GHz will have $T_{\mathrm{sat}} = 0.8$ K. Under these circumstances, to avoid saturation by a broad-band 300-K thermal signal we need at least 20 junctions.

D. Extension to higher frequencies

In this section, we consider the effects of a change in the operating frequency of the SUPARAMP. In particular, we discuss the high-frequency limit of this amplifier. As a first step we can scale the external circuitry with frequency such that the transmission line parameters are unchanged. Once this is done, the operating frequency appears in the final equations of Sec. II only through the dimensionless parameters α_0 and ξ. The apparent dependence of α_0 upon frequency through the relationship $\alpha_0 = 2ev_0/\hbar\omega_0$, however, is not real. In fact, α_0 is an internal variable which is determined by the incident pump power and the other parameters in Eq. (22). Once ξ is specified, α_0 is independent of frequency. Therefore we need consider the frequency dependence of ξ alone.

One method of making a transition to higher frequency is to allow ξ to increase proportionally with the frequency. We then must adjust g to smaller values in order to maintain high gain. (Note that this implies a reduction in the fractional bandwidth and power width.) But we can see from Fig. 6 that there is a largest ξ ($\xi \approx 0.5$), beyond which we cannot operate with high gain. Since at low temperatures

$$I_J R_J \approx 2\Delta/e, \tag{32}$$

where Δ is the zero-temperature energy gap, the condition $\xi \lesssim 0.5$ merely states that we must operate at a frequency less than the gap frequency $\omega_g = 2\Delta/\hbar$ of the superconductor we are using. Conversely, Fig. 6 indicates that this parametric amplifier should indeed yield high gain at frequencies up to about the gap frequency.

Another method of making a transition to higher frequency is to increase the $I_J R_J$ product proportionally with the frequency, so as to keep ξ constant. In this case, we must maintain the same value of g in order to have high gain. We are thus operating at the same point in (g, ξ) space, and would therefore except the same fractional bandwidth and power width. This can be done, for example, by lowering the operating temperature to increase I_J. However, Eq. (32) places an upper limit on $I_J R_J$. To increase the $I_J R_J$ product further, we must use a superconductor with a larger energy gap.

IV. PRELIMINARY EXPERIMENTS

We have observed parametric amplification in a series of unbiased Josephson junctions. Our preliminary experiments at 10 GHz verify all of the qualitative and many of the quantitative predictions of the theory of this paper. We are currently preparing a more extensive series of experiments at 30 GHz,[13a] with the goal of designing an amplifier for radio astronomy. In this section, we discuss those experiments relevant to this theory. For a detailed discussion and analysis of the experiments see Refs. 2 and 14.

The active element in these experiments consists of 80 closely spaced series-connected microbridges,[15] fabricated from a 1000-Å tin film. The dc properties of these junctions were not measured. Two cofabricated single junctions were measured to have normal-state resistance of about 1.4 Ω and an inferred zero-temperature critical current of about 1 mA. The geometrical capacitance of these junctions is known to be negligible.[16]

This string of junctions is incorporated into the upper conductor of a broad-band 50-Ω microstrip transmission line. The junctions have no dc bias. Our operating frequency is near 10 GHz. Termination is accomplished by means of an open circuit at a distance of $\frac{1}{4}\lambda_g$ after the junctions, equivalently placing them in shunt across the transmission line at 10 GHz and all odd harmonic frequencies. Immediately before the junctions there is a tuning stub consisting of a $\frac{1}{12}\lambda_g$ open-ended strip attached at a right angle to the main stripline (see Fig. 8). The purpose of this stub is to short out the third-harmonic frequencies. The tuning stub corresponds to a reduced termination susceptance of $b_T = 1/\sqrt{3} \approx 0.577$ at the pump frequency. A circulator was used to separate the source and the load, both represented by G_0 in Fig. 1.

The general temperature dependence of the effects which we have observed is qualitatively as predicted

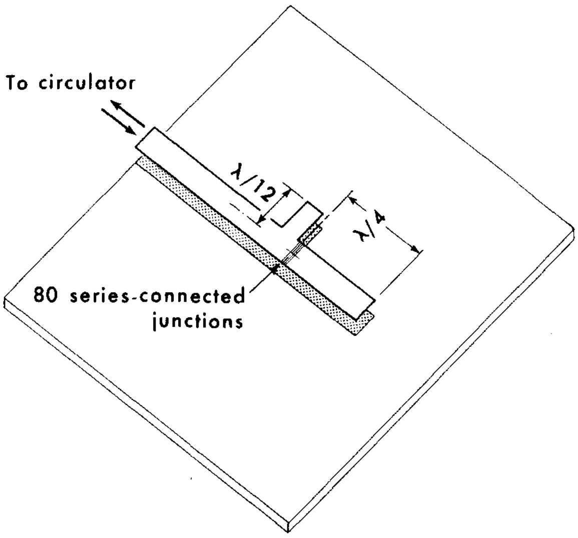

FIG. 8. Microstrip circuitry.

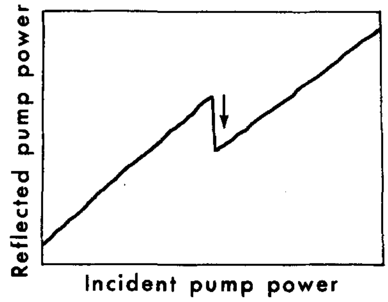

FIG. 9. Chart-recorder trace of reflected pump power plotted against incident pump power, illustrating a glitch. The scale is linear and the incident pump power is on the order of 3 μW.

by the theory. Above a certain temperature we do not see net gain. As the temperature is decreased the gain peak grows and becomes narrower. At still lower temperatures there are glitches in both the gain and the reflected pump power. A chart-recorder trace of a glitch in reflected pump power is shown in Fig. 9.

We found it necessary to operate at a temperature of about 2.5 K, well below the critical temperature of tin, in order to have high gain. At this temperature we have taken careful data. The experimental pump return loss, idler conversion ratio, and signal gain are shown in Fig. 10 as functions of the incident pump power. This is the most complete set of such data taken to date, and has not been processed in any way.

The smooth curves in Fig. 10 are a five-parameter fit to the experimental curves. The values of these parameters for best fit are $I_J = 111$ μA, $R_J = 0.42$ Ω, $N = 19$ junctions, $b_T = 0.556$, and double-pass transmission line loss = 2.3 dB. Of these parameters, the first three characterize the active element, the Josephson junctions, while the last two characterize passive elements of the external circuitry. We assume $\zeta = -1$. Only the signal gain curve is fit. Once this is accomplished, both the pump return loss and the conversion ratio follow directly from the equations without further adjustment of the parameters. The fitting procedure for the signal curve is in essence as follows: Using Fig. 7 and Eq. (28), the power width of the gain curve implies a value for g. From Fig. 6, we can then find the value of ξ which gives the experimental peak gain. We see from Eq. (26) that the incident power at which high gain occurs determines I_J closely, Given g, ξ, and I_J, we find R_J and N. In independent experiments, the double-pass transmission line loss was inferred to be 2.0 ± 0.5 dB. Furthermore, the third-harmonic shorting stub was designed so that $b_T = 0.577$. The values of these two parameters for best fit were found by small variations around their expected values.

We believe this fit to be a reasonable characterization of our experimental situation. Even though the I-V characteristics of cofabricated junctions imply a critical current of about 500 μA at our operating temperature (2.5 K), there are several indications that our critical current is indeed on the order of 100 μA. A pump power -38 dBm will drive a current of about 80 μA in a 50-Ω transmission line. For pronounced nonlinear effects and in particular for high gain, the currents induced in the Josephson element by the pump must be on the order of its critical current. This rough estimate is verified by the pump power required for infinite nonreentrant gain [Eq. (26)], only somewhat less than $\frac{1}{2}I_J^2 Z_0$. In an auxillary experiment, we have measured the small-signal return loss as a function of temperature. This experiment, also done at 10 GHz, implies that $I_J = 105 \pm 10$ μA at 2.5 K. There is no way that the results of this simple experiment, in which the junctions

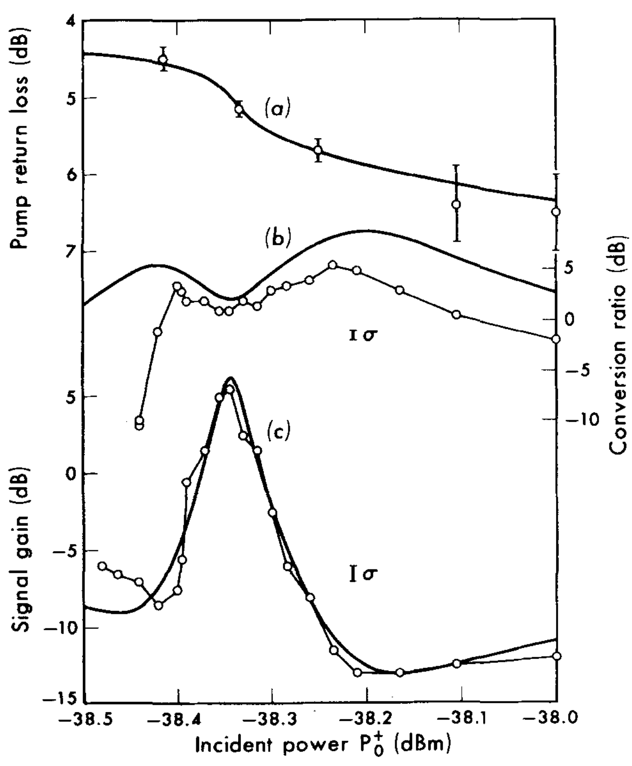

FIG. 10. (a) Pump return loss, (b) conversion ratio, and (c) signal gain plotted against incident pump power. The smooth curves are a simultaneous five-parameter fit to the experimental points. These parameters are $I_J = 111$ μA, $R_J = 0.42$ Ω, $N=19$ junctions, $b_T = 0.556$, and double-pass transmission line passive loss = 2.3 dB. We have assumed $\zeta = -1$.

respond linearly, can be explained with a very much higher critical current. Both of these considerations, indicating a critical current of about 100 μA, are quite general and do not depend upon the detailed analysis of the theory.

There is strong evidence that the effective value of g for our experiment is considerably less than unity ($g = 0.16$ for the theoretical curves in Fig. 10) rather than the originally expected $g = 2.24$ (80, 1.4-Ω junctions in a 50-Ω line). First, for $g > 1$ the signal gain and the conversion ratio curves are smooth; for $g < 1$ they develop characteristic wiggles [compare Figs. 2(b) and 2(c) with Figs. 4(b) and 4(c)] as in the experimental curves. Second, the small power width of the signal gain curve implies $g \ll 1$. Third, if g were much closer to unity the pump return loss dip should be deeper [compare Fig. 10(a) with Figs. 2(a) and 4(a)].

The $I_J R_J$ product from the fitted parameters is about 50 μV, only 0.06 of what we would expect *a priori*. To operate at high frequencies we must resolve this discrepancy.

We speculate as to the reasons for these discrepancies.

(1) We have considered the junctions to be identical, whereas in any real series of junctions there will be some spread in junctions characteristics. Only a fraction of the total number of junctions may be active in producing nonlinear effects. Similarly, the effective values of I_J and R_J for the nonlinear process may differ from those corresponding to the linear process.

(2) In junctions similar to those used in this experiment, we have measured I_J and R_J near the critical temperature from I-V characteristics. These measurements may not be directly applicable. In particular, the normal-state resistance as measured from the Ohmic region of the I-V characteristic might be very different from the quasiparticle resistance R_J at low temperature at $I = 0$ and $V = 0$.

(3) There is evidence[17] that in the very smallest microbridges the critical current increases very slowly as the temperature is decreased, as opposed to the rapid increase seen in all other junctions. This could account for the anomalously small value inferred for I_J at low temperature.

The FWHM bandwidth of this device was roughly measured to be 1 GHz.[2,14] This is in good agreement with the predictions of Fig. 7, for the inferred value of g.

V. CONCLUDING REMARKS

In this paper, we have presented an analysis of a new type of amplifier, the SUPARAMP, in an attempt to elucidate its properties and to understand its various regimes of operation. Some qualitatively new features not present in the usual varactor parametric amplifier are found: (i) the inversion symmetry of the active element, implying the absence of the sum, the difference, and the second-harmonic frequencies, (ii) a maximum in the gain as the pump power is increased monotonically, (iii) the reentrance and glitching, (iv) the possibility of achieving high gain without resonating out the self-coupling reactance. Some quantitatively new features are the small required pump power and the large bandwidth.

The cutoff frequency of a varactor paramp is determined by parasitic elements, usually the series resistance and the combination of diode and stray capacitance. However, for the simple gometry of the SUPARAMP, the effects of such parasitic elements should be negligible. The high-frequency cutoff of this device is the gap frequency of the superconducting material used for the active element. For instance, for tin this frequency is 280 GHz and for NbN, 1200 GHz.

We have not analyzed the noise properties of the SUPARAMP. But in common with other parametric amplifiers, the active element is basically a lossless nonlinear reactance. Since this device is of necessity at liquid-helium temperatures, any residual losses are cold losses. We therefore expect a very low noise temperature for the SUPARAMP (see Ref. 2).

APPENDIX: HARMONIC CALCULATION

The total current through the bare Josephson elements at all frequencies of interest is

$$i = I_J \sin(\phi_0 + \phi_{30} + \phi_{50} + \phi_w), \tag{A1}$$

where

$$\phi_w = \sum_{i=1}^{6} \alpha_i \sin(\omega_i t + \theta_i).$$

The subscripts 30 and 50 refer to the third and fifth harmonics of the pump frequency ω_0, respectively. We have indexed the weak waves $i = 1$–6, at frequencies $n\omega_0 \pm (\omega_0 - \omega_1)$, $n = 1, 3$, and 5, in the order of ascending frequency (assuming for convenience $\omega_1 < \omega_0$). We assume voltages v_k analogous to Eq. (7) and define α_k and ϕ_k analogous to Eq. (8).

To solve the strong-wave problem, we ignore ϕ_w in Eq. (A1), assume that α_{30} and α_{50} are small, as discussed in the text, and expand $\sin\phi_0$ in terms of Bessel functions. The amplitudes of the currents flowing through the bare Josephson elements at ω_0, $3\omega_0$, and $5\omega_0$ are, in complex notation,

$$i_0/I_J = -j(2J_1 + J_2\alpha_{30} - J_4\alpha_{30}^* + J_4\alpha_{50} - J_6\alpha_{50}^*), \tag{A2}$$

$$i_{30}/I_J = -j(2J_3 + J_0\alpha_{30} - J_6\alpha_{30}^* + J_2\alpha_{50} - J_8\alpha_{50}^*), \tag{A3}$$

$$i_{50}/I_J = -j(2J_5 + J_2\alpha_{30} - J_8\alpha_{30}^* + J_0\alpha_{50} - J_{10}\alpha_{50}^*), \tag{A4}$$

where the common argument of all Bessel functions is α_0. The circuit equations at these three frequencies for the junctions embedded in the external network are

$$i_0/I_J + \xi(1 + jgb_T)\alpha_0 = I_0/I_J, \tag{A5}$$

$$i_{30}/I_J + 3\xi(1 + g_{30} + jg_{30}b_{T30})\alpha_{30} = 0, \tag{A6}$$

$$i_{50}/I_J + 5\xi(1 + g_{50} + jg_{50}b_{T50})\alpha_{50} = 0. \tag{A7}$$

In Eqs. (A6) and (A7), we have used dimensionless parameters analogous to those in the text, but evaluated at frequencies $3\omega_0$ and $5\omega_0$. Equations (A3), (A4), (A6), and (A7) are linear and can be solved straightforwardly for α_{30} and α_{50}, given α_0. The incident and reflected pump powers, from Eqs. (A2) and (A5), are

$$P_0^{\pm}=\tfrac{1}{8}I_J^2 Z_0\left|2J_1+J_2\alpha_{30}-J_4\alpha_{30}^* + J_4\alpha_{50}-J_6\alpha_{50}^*+j(1\pm g+jgb_T)\xi\alpha_0\right|^2.$$

Knowing α_0, α_{30}, α_{50}, we now proceed to solve the weak-wave problem. Assuming $\alpha_i \ll 1$ in Eq. (A1), the total weak-wave current through the bare Josephson elements is

$$i_w = I_J\left(\sum_{i=1}^{6}\alpha_i\sin(\omega_i t+\theta_i)\right) \times\left[J_0+2\sum_{l=1}^{\infty}J_{2l}\cos 2l\omega_0 t-2\left(\sum_{l=1}^{\infty}J_{2l-1}\sin(2l-1)\omega_0 t\right) \times[\alpha_{30}\sin(\omega_{30}t+\theta_{30})+\alpha_{50}\sin(\omega_{50}t+\theta_{50})]\right]. \tag{A8}$$

The weak-wave admittance matrix will be a 6×6 matrix. The matrix element Y_{mn} gives the current at the weak-wave frequency ω_m induced by a voltage at the weak-wave frequency ω_n. We first examine the one term of Y_{mn} which does not depend upon α_{30} or α_{50}. There is one and only one such term, for there is one and only one equation relating ω_m to ω_n under our assumptions, namely,

$$\omega_m = s\omega_n + k\omega_0, \tag{A9}$$

where $s=\pm 1$ and k is an integer. $|s|>1$ is not allowed since $\alpha_n \ll 1$. We can see from Eq. (A8) that k must be even. The term we are looking for can be isolated from Eq. (A8):

$$i_m^{(n)}\exp(j\omega t) = -jI_J[\alpha_n\exp(j\omega_n t)-\alpha_n^*\exp(-j\omega_n t)] \times J_k\exp(jk\omega_0 t),$$

where we have defined $i_m^{(n)}$ to be the current at frequency ω_m induced by a voltage at frequency ω_n. Dropping the exponential time dependence by use of Eq. (A9), we see that

$$i_m^{(n)}\exp(j\omega_m t) = -jI_J J_k\alpha_n, \quad \text{for } m+n \text{ even}$$

$$i_m^{(n)} = +jI_J J_k\alpha_n^*, \quad \text{for } m+n \text{ odd}.$$

This equation gives only one term of $i_m^{(n)}$, the term which does not depend upon α_{30} and α_{50}. Under our assumption that α_{30} and α_{50} are small, there are four more terms which must be considered, given by the frequency relations

$$\omega_m = s\omega_n + (k-qp)\omega_0 + q\omega_{p0},$$

where $q=\pm 1$, $p=3$ or 5, and s and k are the same as in Eq. (A9). These other four terms contributing to $i_m^{(n)}$ are found by following a procedure similar to the above. Including all terms, we have

$$i_m^{(n)} = -\tfrac{1}{2}j\alpha_n I_J(2J_k+J_{k-3}\alpha_{30}-J_{k+3}\alpha_{30}^*+J_{k-5}\alpha_{50}-J_{k+5}\alpha_{50}^*)$$

for $m+n$ even,

$$i_m^{(n)} = +\tfrac{1}{2}j\alpha_n^* I_J(2J_k+J_{k-3}\alpha_{30}-J_{k+3}\alpha_{30}^*+J_{k-5}\alpha_{50}-J_{k+5}\alpha_{50}^*)$$

for $m+n$ odd.

There is an arbitrary convention involved in defining the elements of the admittance matrix. As in the nonharmonic weak-wave calculation, we define

$$Y_{mn} = i_m^{(n)}/v_n, \quad \text{for } m+n \text{ even}$$

$$Y_{mn} = i_m^{(n)*}/v_n, \quad \text{for } m+n \text{ odd}.$$

Considering the circuit in which the Josephson elements are embedded, the current-voltage equations at the various frequencies of interest are

$$I_m' = \sum_{n=1}^{6} Y_{mn}' V_n'/N + (G_J/N + Y_{Tm}')V_m', \tag{A10}$$

where $I_1 = Y_1 V_1$ and $I_m = -G_m V_m$ for $m\neq 1$. We have defined a set of primed quantities:

$I_n' = I_n$, n odd $\quad V_n' = V_n$, n odd

$I_n' = I_n^*$, n even; $\quad V_n' = V_n^*$, n even;

$Y_{mn}' = Y_{mn}$, n odd $\quad Y_{Tn}' = Y_{Tn}$, n odd

$Y_{mn}' = Y_{mn}^*$, n even; $\quad Y_{Tn}' = Y_{Tn}^*$, n even.

Using a digital computer, the set of simultaneous linear equations (A10), $m=2$–6, can be solved for V_m'/V_1. The input admittance at the signal frequency, Y_1, follows immediately from Eq. (A10), $m=1$. The signal gain is given by Eq. (15) but with G_0 replaced by G_1, and the conversion ratio at frequency ω_n is given by

$$C_n = 4G_1G_n\left|\frac{V_n^*/V_1}{Y_1-G_1}\right|^2.$$

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